

Daniel Crocker

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EDUCATION

2021	University of California, San Diego <i>Chemistry and Biochemistry, PhD</i>	La Jolla, CA
2014	University of Connecticut <i>Chemistry, B.S. with Mathematics minor</i>	Storrs, CT

RESEARCH EXPERIENCE

2024 - 2026	NSF Postdoctoral Fellow in Atmospheric and Geospace Sciences Earth & Planetary Sciences Department, Harvard University Project: <i>Triple Oxygen Isotope Investigations of Organic Aerosol Oxidation</i> Primary Advisor: David Johnston Secondary Advisor: Meredith Hastings
2022 - 2024	Harvard Origins of Life Initiative Postdoctoral Scholar Earth & Planetary Sciences Department, Harvard University Project: <i>Development and Application of Novel Methodology for Triple Oxygen Isotope Measurements of Meteoritic Organic Matter</i> Primary Advisor: David Johnston Secondary Advisor: Ann Pearson
2014 - 2021	Ph.D. in Atmospheric Chemistry UC San Diego, Department of Chemistry and Biochemistry Center for Aerosol Impacts on Chemistry of the Environment (CAICE) <i>Ph.D. Thesis: Investigations into Biological Influences on the Carbon Isotopic Composition ($\delta^{13}\text{C}$) of Nascent Sea Spray Aerosol and Ocean-Aerosol Transfer of Organic Material</i> Advisor: Mark Thiemens
2011 - 2014	B.S. in Chemistry University of Connecticut, Department of Chemistry <i>Honors Thesis: Photodegradation of Dibenzothiophene Using Titania Catalysts: Applications in Fuel Desulfurization</i> Advisor: Steven Suib

Lead author publications:

- (1) **Crocker, D. R.**; Alexander, C. M.O'D.; Thiemens, M. H.; Johnston, D. T. Meteoritic Organic Matter Records Primitive Oxygen Reservoirs of the Solar System. *Proc. Natl. Acad. Sci.*, 2026, 123 (40), e2605307123.
- (2) **Crocker, D. R.**; Sutherland, K. M.; Freudenberg, B.; Alonso, K.; Ward, C.; Pearson, A.; Johnston, D. T. Tracking photo-oxidation reactions of aquatic organic matter using triple oxygen isotopes. *ACS Earth Space Chem.*, 2025, 9 (7), 1780–1790.
- (3) **Crocker, D. R.**; Itahashi, S.; Hattori, S.; Boothby, C.; Valdes, D.; Zhou, Q.; Cao, R.; Le, K.; Jackson, T.; Thiemens, M. H. Isotopic examination of variations in sulfur(IV) oxidation pathways during the COVID shutdown. *ACS ES&T Air*, 2025, 2 (12), 2954-2965.
- (4) **Crocker, D. R.**; Kaluarachchi, C.; Cao, R.; Morris, C.K.; Franklin, E. B.; Dinasquet, J.; Amiri, S.; Petras, D.; Nguyen, T.; Torres, R. R.; Martz, T. D.; Malfatti, F. M.; Goldstein, A. H.; Tivanski, A. V.; Prather, K. A.; Thiemens M. H. Isotopic Insights into Organic Composition Differences between Supermicron and Submicron Sea Spray Aerosol. *Environ. Sci. Technol.*, 2022, 56 (14), 9947-9958.
- (5) **Crocker, D. R.**; Deane, G. B.; Cao, R.; Santander M. V.; Morris, C. K.; Mitts, B.A.; Dinasquet, J.; Malfatti, F. M.; Prather, K. A.; Thiemens, M. H. Biologically Induced Changes in the Partitioning of Submicron Particulates between Bulk Seawater and the Sea Surface Microlayer. *Geophys. Res. Lett.*, 2022, 49, e2021GL094587.
- (6) **Crocker, D. R.**; Hernandez, R.; Huang D. H.; Pendergraft, M. A.; Cao, R.; Dai, J.; Morris, C. K.; Deane, G. B.; Prather, K. A.; Thiemens, M. H. Biological Influence on $\delta^{13}\text{C}$ and Organic Composition of Nascent Sea Spray Aerosol. *ACS Earth Space Chem.*, 2020, 4 (9), 1686-1699.

Co-authored Publications:

- (1) Hong, Y.; Zhu, L; **Crocker, D. R.**; et al., “Triple Oxygen Isotope Analysis of Methanesulfonate Using the SO_3^- Fragment in ESI-Orbitrap-MS.” *Analytical Chemistry*, 2025, 97 (27), 14339-14348.
- (2) Jiang H. J.; Underwood T. C.; Bell J. G.; Lei J.; Gonzales J. C.; Emge L.; Tadese L. G.; El-Rahman M. K. A.; Wilmouth D. M.; Brazaca L. C.; Ni G.; Belding L.; Dey S.; Ashkarran A. A.; Nagarkar A.; Nemitz M. P.; Cafferty B. J.; Sayres D. S.; Ranjan S.; **Crocker D. R.**; Anderson J. G.; Sasselov D. D.; Whitesides G. M. Mimicking lightning-induced electrochemistry on the early Earth. *Proc. Natl. Acad. Sci.*, 2024, 121, e2400819121.
- (3) Selden C. R.; LaBrie, R.; Ganley, L. C.; **Crocker, D. R.**; Peleg, O.; Perry, D.; Reich, H. G.; Sasaki, M.; Thibodeau, P. S.; Isanta-Navarro, J. Is our understanding of aquatic ecosystems sufficient to quantify ecologically driven climate feedbacks? *Global Change Biology*, 2024, 30 (6), e17351.
- (4) Franklin, E. B.; Amiri, S.; **Crocker, D.**; Morris, C.; Mayer, K.; Sauer, J. S.; Weber, R. J.; Lee, C.; Malfatti, F.; Cappa, C. D.; Bertram, T. H.; Prather, K. A.; Goldstein, A. H. Anthropogenic and Biogenic Contributions to the Organic Composition of Coastal Submicron Sea Spray Aerosol. *Environ. Sci. Technol.* 2022, 56 (23), 16633-16642.
- (5) Sauer, J. S.; Mayer, K. J.; Lee, C.; Alves, M. R.; Amiri, S.; Bahaveolos, C.; Barnes, E. B.; **Crocker, D. R.**; et al., The Sea Spray Chemistry and Particle Evolution Study (SeaSCAPE): Overview and Experimental Methods. *Environ. Sci. Process. Impacts*, 2022, 24 (2), 290-315.

- (6) Kaluarachchi, C. P.; Or, V.W.; Lan, Y.; Madawala, C.K.; Hasenecz, E. S.; **Crocker, D. R.**; et al., Size-Dependent Morphology, Composition, Phase State, and Water Uptake of Nascent Submicrometer Sea Spray Aerosols during a Phytoplankton Bloom. *ACS Earth Space Chem.*, 2022, 6 (1), 116-130.
- (7) Kilgour, D. B., Novak, G. A., Sauer, J. S., Moore, A. N., Dinasquet, J., Amiri, S., Franklin, E. B., Mayer, K., Winter, M., Morris, C. K., Price, T., Malfatti, F., **Crocker, D. R.**, Lee, C., Cappa, C. D., Goldstein, A. H., Prather, K. A., & Bertram, T. H. Marine gas-phase sulfur emissions during an induced phytoplankton bloom. *Atmospheric Chem. Phys.*, 2022, 22 (2), 1601–1613.
- (8) Tumminello, P. R.; James, R. C.; Kruse, S.; Kawasaki, A.; Cooper, A.; Guadalupe-Diaz, I.; Zepeda, K. L.; **Crocker, D. R.**; et al., Evolution of Sea Spray Aerosol Particle Phase State Across a Phytoplankton Bloom. *ACS Earth Space Chem.*, 2021, 5 (11), 2995-3007.
- (9) Angle, K. J.; **Crocker, D. R.**; et al., Acidity across the interface from the ocean surface to sea spray aerosol. *Proceedings of the National Academy of Sciences*, 2021, 118 (2), e2018397118.
- (10) Lin, M.; Biglari, S.; Zhang, Z.; **Crocker, D.**; Tao, J.; Su, B.; Liu, L.; Thiemens, M. H. Vertically uniform formation pathways of tropospheric sulfate aerosols in East China detected from triple stable oxygen and radiogenic sulfur isotopes. *Geophys. Res. Lett.*, 2017, 44 (10), 5187–5196.

Under Review:

- (1) **Crocker, D. R.**; Goranov, A. I.; Pearson, A.; Hatcher, P. G.; Johnston, D. T. Using oxygen isotopes to trace organic matter oxidative degradation via Fenton chemistry. [manuscript in revision for ES&T]
- (2) **Crocker, D. R.**; Kim, H.; Doi, E. K.; Goss, M. B.; Hastings, M. G.; Johnston, D. T. Triple oxygen isotope constraints on organic aerosol oxidation pathways. preprint: doi 10.21203/rs.3.rs-10959350/v1 [manuscript submitted to Science Advances]

RESEARCH PRESENTATIONS

Seminars & Colloquia

- (1) **Crocker, D. R.**: Using triple oxygen isotopes to track organic matter chemistry and cycling in Earth systems, *Brown University Climate & Environment Lunch Bunch Series*, Providence, RI, USA, May 2026
- (2) **Crocker, D. R.**: Oxygen Isotopes of Meteoritic Organic Matter: Insights into Early Solar System Organic Chemistry and Oxygen Reservoirs, *Harvard Origins Federation Chalk Talk*, Cambridge, MA, USA, October 2025
- (3) **Crocker, D. R.**, Freudenberg, B.; Johnston, D. T.: Isotopes as a Tool to Investigate Carbonaceous Aerosol Sources, Atmospheric Chemistry, and Climate Influence, *Salata Institute for Climate Scholar Seminar Series*, Cambridge, MA, USA, March 2024
- (4) **Crocker, D. R.**; Freudenberg, B.; Alonso, K. A.; Sutherland, K. M.; Pearson, A.; Johnston, D. T.: An Innovative Methodological Workflow to Enable Triple Oxygen Isotope Analysis of Organic Matter, *Aerodyne Research Inc. Seminar*, Billerica, MA, USA, August 2023
- (5) **Crocker, D. R.**; Freudenberg, B.; Henkes, G. A.; Mak, J.; Pearson, A.; Johnston, D. T., Thiemens, M. H.: Tracking the Sources, Chemistry, and Climate Impacts of Short-Lived Aerosols and Trace Gases, *Utrecht University Department Seminar*, Utrecht, NL, June 2023
- (6) **Crocker, D. R.** What can oxygen isotopes of carbonaceous chondrites tell us about prebiotic organic chemistry in our Solar System?, *Harvard Origins of Life Initiative Chalk Talk*, Cambridge, MA, USA, May 2023
- (7) **Crocker, D. R.**; Hernandez, R.; Huang D. H.; Pendergraft, M. A.; Cao, R.; Dai, J.; Prather, K. A.; Thiemens, M. H: Carbon Stable Isotopic Characterization of Biological Influences on Nascent Sea Spray Aerosol Chemical Composition, *CAICE Monthly Colloquium*, San Diego, CA, USA, January 2020

Conference Submissions

- (1) **Crocker, D. R.**; Alexander, C.M.O.D; Thiemens, M. T.; Johnston, D. T.: Oxygen isotope evidence that chondrites accreted a common organic component. *Origins of Life Federation Annual Conference*, Lucerne, Switzerland, 2026
- (2) **Crocker, D. R.**; Kim, H.; Doi, E. K.; Goss, M. B.; Hastings, M. G.; Johnston, D. T. Quantitatively constraining organic aerosol oxidation pathways with triple oxygen isotopes. *American Association for Aerosol Research 44th Annual Conference*, Pasadena, CA, 2026
- (3) **Crocker, D. R.**; Alexander, C.M.O.D; Thiemens, M. T.; Johnston, D. T.: Oxygen isotope evidence that chondrites accreted a common organic component. *Lunar and Planetary Science Conference*, Woodlands, TX, USA, 2026
- (4) **Crocker, D. R.**; Alexander, C.M.O.D; Thiemens, M. T.; Johnston, D. T.: Meteoritic Organic Matter Records Primitive Oxygen Reservoirs of the Solar System. *Origins of the Solar System Gordon Research Conference*, South Hadley, MA, USA, 2025
- (5) **Crocker, D. R.**; Freudenberg, B.; Pearson, A.; Alexander, C.; Johnston, D. T.: Triple Oxygen Isotope Evidence that Insoluble Organic Matter in Carbonaceous Chondrites Shares a Related Chemical History with the Anhydrous Minerals. *Origins of Life Federation Annual Conference*, Cambridge, UK, 2024
- (6) **Crocker, D. R.**; Freudenberg, B.; Sutherland, K. M.; Keller, K.; Blewett, J.; Alonso, K. A.; Ward, C.; Henkes, G.A.; Mak, J.; Pearson, A.; Johnston, D. T.: Development and Environmental Applications of a Methodological Workflow for Triple Oxygen Isotope Analysis of Organic Matter, *AGU Fall Meeting*, San Francisco, CA, USA, 2023
- (7) **Crocker, D. R.**; Boothby, C.; Valdes, D.; Zhou, Q.; Cao, R.; Le, K.; Jackson, T.; Thiemens, M. H: An Isotopic Perspective on Changes in Sulfate Sources during the COVID-19 Shutdown in Southern California, *ACS Fall Meeting*, Chicago, IL & Virtual, USA, 2022
- (8) **Crocker, D. R.**; Kaluarachchi, C.; Cao, R.; Morris, C.K.; Franklin, E. B.; Dinasquet, J.; Amiri, S.; Petras, D.; Nguyen, T.; Torres, R. R.; Martz, T. D.; Malfatti, F. M.; Goldstein, A. H.; Tivanski, A. V.; Prather, K. A.; Thiemens M. H: Carbon Isotopic Investigations of Compositional Differences between Submicron and Supermicron Sea Spray Aerosol, *ACS Spring Meeting*, Virtual, 2021
- (9) **Crocker, D. R.**; Hernandez, R.; Huang D. H.; Pendergraft, M. A.; Cao, R.; Dai, J.; Morris, C. K.; Deane, G. B.; Prather, K. A.; Thiemens, M. H: Biological Influence on $\delta^{13}\text{C}$ and Organic Composition of Nascent Sea Spray Aerosol, *AGU Fall Meeting*, Virtual, 2020
- (10) **Crocker, D. R.**; Hernandez, R.; Pendergraft, M. A.; Cao, R.; Dai, J.; Prather, K. A.; Thiemens, M. H: Carbon Isotopic Characterization of Biological Influences on Sea Spray Aerosol Chemical Speciation, *ACS Fall Meeting*, San Diego, CA, USA, 2019
- (11) **Crocker, D. R.**; Pahalagedara, L.; Suib, S.: Photodegradation of Dibenzothiophene using Titania Catalysts. *ACS Connecticut Valley Section Undergraduate Symposium*, Amherst, MA, USA, 2014

FUNDING & AWARDS

Total: \$ 828,026

- 2025 NASA Emerging Worlds research grant (two-year, \$493,526)
- *Project Title: The Triple Oxygen Isotope Composition of Meteoritic Organic Matter; Insights into its Origin and Chemical Evolution*
- 2024 NSF-AGS Postdoctoral Fellowship (two-year, \$202,000)
- *Project Title: Triple Oxygen Isotope Investigations of Organic Aerosol Oxidation*
- 2023 Harvard Dean's Competitive Fund for Promising Scholarship (six-month, \$43,000)
- 2021 Harvard Origins of Life Initiative research grant (one -year, \$89,000)
- *Project Title: Development and Fabrication of Instrumental Analysis System for Triple-Oxygen Isotope Measurements on Organic Molecules*
- 2020 CAMECA Instruments Travel Grant (\$500)

PROFESSIONAL TEACHING EXPERIENCE

UC San Diego, Chemistry and Biochemistry

Instructor of Record, General Chemistry II, Summer 2020 (45 students, virtual)

Graduate Teaching Assistant for 25 academic quarters

Environmental Chemistry II: (Winter 2020, Winter 2021)

Basic Chemistry Laboratory: (Fall 2014, Fall 2015, Fall 2016, Fall 2017, Fall 2019, Fall 2020)

General Chemistry II: (Winter 2015, Spring 2017, SummerII 2017, Winter 2018, Fall 2018, Winter 2019, Spring 2020, SummerI 2021)

General Chemistry III: (Spring 2015, Spring 2016, SummerI 2017, Spring 2018, SummerI 2018, SummerII 2018, Spring 2021, SummerII 2021)

Intro to Organic Chemistry: (Winter 2017)

Teaching Accolades

2020 Summer Graduate Teaching Scholar (SGTS)

2020 UCSD Excellence in Teaching Award

2018-19 Senior Teaching Assistant

Pedagogical Training

2014 "Teaching Methods in Chemistry" course on chemistry teaching pedagogy

2020 "Introduction to College Teaching" course on university level pedagogical instruction

2023 Harvard Museum of Science Education Partners program on K-12 science communication

MENTORSHIP & ADVISING EXPERIENCE

2022-present Mentor for Geosciences Education & Mentorship Support (GEMS) Program
- *Advising undergraduate students in selecting and applying to graduate school*

2026-present Ashimata Prajapati, Undergraduate Researcher, Harvard University, Johnston Lab

2024-2026	Emily Doi, Undergraduate Researcher, Harvard University, Johnston Lab
2022-2024	Ben Freudenberg, Undergraduate Researcher, Harvard University, Johnston Lab -Current Status: Research Fellow at Environmental Protection Agency
2022-2023	Kimberly Alonso, Arizona State University, Origins of Life Summer Researcher, Johnston Lab -Current Status: Astrophysical Instrumentation Research Intern, Harvard University
2020-2021	Dagoberto Valdes, Undergraduate Researcher, UC San Diego, Thiemens Lab
2020-2021	Christian Boothby, Undergraduate Researcher, UC San Diego, Thiemens Lab -Current Status: Ph. D. Student, UC San Diego
2018-2021	Ruochen (Charlie) Cao, Undergraduate Researcher, UC San Diego, Thiemens Lab -Current Status: R&D at Genentech Inc.
2018-2019	Jiayin (Jayne) Dai, Undergraduate Researcher, UC San Diego, Thiemens Lab -Current Status: Ph. D. Student, UC Davis
2018-2019	Jiawei Yin, Undergraduate Researcher, UC San Diego, Thiemens Lab -Current Status: M.S. Student, Yale University
2015-2017	Ritchie Hernandez, NSF-REU Undergraduate Researcher, UC San Diego, Thiemens Lab -Current Status: Ph. D. Student, UC San Diego

SERVICE AND OUTREACH ACTIVITIES

Committee Experience

2026	Session Moderator for Lunar and Planetary Science Conference <i>Session Title: "Chondrites as Records of Solar System Processes"</i>
2025	Co-convener for Atmospheric Sciences Session at 2025 AGU Fall Meeting <i>Session Title: "Isotopes of the atmospheric components: laboratory investigation, field observation, modeling, and remote sensing"</i>
2024	Co-convener for Atmospheric Sciences Session at 2024 AGU Fall Meeting <i>Session Title: "Isotopes of the atmospheric components: laboratory investigation, field observation, modeling, and remote sensing"</i>
2023	Co-convener for Union Session at 2023 AGU Fall Meeting <i>Session Title: "40 Years of Theoretical Advances in Isotope Effects and Applications in Earth, Atmospheric, and Planetary Sciences"</i>
2019	Graduate Student Symposium Planning Committee for ACS Fall Meeting <i>Symposium Title: "From Oceans to Clouds: The Environmental Chemistry of Water"</i>
2017	Student convener for 34 th Informal Symposium of Kinetics and Photochemical Processes in the Atmosphere (ISKPPA)
2015-2021	Graduate Student Member of Academic Integrity Review Board at UC San Diego

Peer Reviewer

Proceedings of the National Academy of Sciences, Earth and Planetary Science Letters, Journal of Geophysical Research: Atmospheres, Journal of the American Chemical Society, MDPI: Atmosphere, NSF CAREER Grant

Educational Outreach

- 2023-present Harvard Museums of Science Education “Virtual Scientist in a Classroom” program
-Conduct virtual Q&A sessions with middle and high school classrooms on Earth science research
- 2024 Harvard Museums of Science and Culture Spotlight Talk
Title: “Space Rocks and the Origin of Life”
- 2022-present Question writer for Middle School and High School National Science Bowl® competition
- 2020-2021 CAICE Clear program mentor for middle school students to develop and investigate a hypothesis on aerosol science and present the results at a university-sanctioned symposium
- 2015-2017 Judging panelist for CAICE Clear program poster symposium
- 2015, 2017 Hosted chemistry booth at the San Diego Science Exposition
- 2015 Hosted chemistry booth at Miramar College Science Exposition